

The JRPG Speedrun Exploits Gauntlet

Description

Undertale is one of the most influential games in the Japanese Role Playing Game (JRPG) genre released in 2015 developed by American indie developer Toby Fox. The game released quietly, but many who invested in its kickstarter project were blown away by Undertale's masterful execution of its characters, story, music, and atmosphere. Shortly after, the game exploded in popularity when the most influential content creators at the time began to discover Undertale.

While Toby Fox was considered masterful at writing stories and music, programming was a skill he was not the best at. Undertale, developed under the GameMaker engine, is Toby Fox's first fully developed video game. He made several oversights which allowed for several game breaking glitches and exploits to be discovered. The most notable glitch found in Undertale is the "Punch Card Exploit" (PCE), which is a method of regaining movement during situations where you are not supposed to move. This allowed players to go into pixel-perfect positions that triggered cutscenes, do a PCE to regain movement, then walk past the cutscene to skip it. These exploits transformed Undertale to one of the most popular speed running title to the point that its speedrun.com leaderboard verification queue still sits past one hundred.

This project is intended to study Undertale's flaws in its mechanics to understand how the PCE and other types of known exploits work. Then, knowing how and why these exploits work, the plan is to purposely build a short JRPG game level and implement these flaws for players to use them. The goal of this game is to beat the level as fast as possible. Players would be encouraged to use these exploits to save time. As the game would be developed for a speed running audience, there will be many quality of life features like an in-game timer, in-game leaderboard, a tracker of exploits used, and a practice environment.

The Project's Value

Currently, Undertale's Any% glitched category's world record sits at 51 minutes, 26 seconds, and 383 milliseconds. This run included over a hundred uses of glitches and exploit to saves several minutes over beating the game without the use of glitches. The project's goal is to purposely implement several instances where each of these exploits

could be used over one entire level. This project's value comes from delivering the Undertale Any% experience in one condensed level within a 5-10 minute time period. It could be considered a demo of the Undertale Any% glitched category. Learning the speed run of Undertale is a massive hurdle, and this project aims to make that hurdle easier to pass by having the player learn and practice speed running techniques in a condensed experience.

Skills

- Problem solving skills.
- Interpersonal skills for a team environment or the ability to work fully independently
- Familiarity with common 2D game development frameworks and engines (GameMaker, LibGDX, Godot, etc.).
- Debugging knowledge in GameMaker products.
- Understanding of what makes a video game fun.
- Surface level understanding of why and how Undertale works as a speed game.
- Deep level knowledge of Undertale's events and variable structure and functionality.
- Knowledge of the essential game developer concepts to develop and animate on a 2D world.
- Individual and team project management skills.
- Deep level understanding of data structures and algorithms to produce an optimized working product.
- Knowledge of typical game development languages: C/C++, C#, Java, and Python.
- Familiarity of source control products like Git.

Personal Requirements

- Deep level study of Undertale's mechanics, systems, engine, and exploits to replicate in our project.
- Determine the appropriate game engine or framework to develop a small 2D game within a 6-8 week period.

- Design a short 5-10 minute game level that incorporates the following Undertale exploits: punch card exploit, text choicer overflow, and wrong warping (overflow or punch card).
- A responsive input handling system that feels like Undertale's Linux version.
- A scene system that containerizes the title screen, game over scene, results screen, and the game section.
- A simple RPG battle system that incorporates satisfying rhythmic elements to reward skillful gameplay.
- A sound system to deliver a welcoming, yet energizing ambience and atmosphere.
- A cutscene and dialogue system that can be finished faster using exploits or rhythmic mashing.
- Speed running quality of life features like an in-game timer and recording of one's personal bests.
- Tutorial features that teach the player how to execute Undertale exploits to save time. This could be in the form of a tutorial stage and a practice environment.

Undertale Glitch Showcase

The following two links show the difference of skipping an Undertale boss with and without the use of a glitch. The purpose is to show the basic technique of a punch card exploit.

Skipping Muffet boss without Glitches

<https://youtu.be/WRYlNXuaxlc>

Skipping Muffet boss with the Punch Card Exploit

<https://youtu.be/vvRQuAojtBk>